Skills Summary

End Behavior and Multiplicity

Use this reference while completing your worksheet or when studying.

Skill 1 — End behavior: degree and lead coefficient

Rule: For large $|x|$ the leading term dominates every other term, so only two facts matter. **Degree parity** says whether the ends agree (even: agree; odd: disagree). **Sign of the leading coefficient** says which way the right end goes (positive: up; negative: down).

Degree	Lead. coeff.	Left	Right
even	positive	up	up
even	negative	down	down
odd	positive	down	up
odd	negative	up	down

Example: End behavior of $f(x) = -4x^3 + x$ as $x \to +\infty$.
Step 1. Only the leading term matters at the ends: odd degree with a negative coefficient, so the ends disagree and the right one falls.
Step 2. $f(x) \to \boldsymbol{-\infty}$ as $x \to +\infty$ (and $+\infty$ on the left)

Try it: What does $f(x) = 5x^4 - 2x$ do as $x \to -\infty$? check: $+\infty$

Skill 2 — Zeros and what multiplicity does there

Rule: Factored form hands you the zeros: set each factor to zero. The exponent on that factor is its *multiplicity.*

- **Odd** multiplicity: the graph **crosses** the axis (a multiplicity of three crosses flattened out).
- **Even** multiplicity: the graph **touches and turns back**, and the sign does not change.

Degree = the sum of the multiplicities, so the zeros also cap how many turns the curve can have.

Example: Zeros of $f(x) = (x-4)(x+1)^2$.
Step 1. The function is already factored, so read each factor and its exponent rather than expanding anything.
Step 2. $x - 4 = 0$ gives $x = 4$ (multiplicity one, crosses); $(x+1)^2 = 0$ gives $x = -1$ (multiplicity two, touches): $\boldsymbol{x = 4}$ (crosses)**,** $\boldsymbol{x = -1}$ (touches)

Try it: List the zeros of $f(x) = x^2(x-5)$ and say what the graph does at each.

check: $\boldsymbol{x = 0}$ touches, $\boldsymbol{x = 5}$ crosses

Skill 3 — Signs, the y-intercept, and the sketch

Rule: Between consecutive zeros the sign of a polynomial cannot change, so *one* test value decides a whole interval. Order of work for a sketch: end behavior $\to$ zeros with multiplicities $\to$ $f(0)$ $\to$ one test value per remaining interval $\to$ join the pieces with a smooth curve.

Example: Find the y-intercept of $f(x) = (x+1)(x-2)^2$ and give the sign of f just right of $x = 0$.

Step 1. The y-intercept is the test value at $x = 0$, which also fixes the sign on the whole interval between the zeros -1 and 2.

Step 2. $f(0) = (1)(-2)^2 = \mathbf{4}$, positive — so $f > 0$ across $(-1, 2)$

Try it: Find $f(0)$ for $f(x) = (x-3)(x+2)^2$. check: $\mathbf{-12}$